Data Objects and Attribute Types

1 Data Objects

A dataset is fundamentally a collection of **data objects**.

Definition

data object represents a real-world **entity**—such as a customer, patient, or student—that is described by a set of attributes.

Example

- **Sales database:** customers, products, transactions
- **Medical database:** patients, diagnoses, treatments
- **University database:** students, professors, courses

Alternative Terminology

Data objects may also be referred to as:

- Samples, instances, or examples
- Data points
- Tuples (database rows)

Database Concept	Data Mining Term
Row	Data Object / Tuple
Column	Attribute / Feature

2 Attributes

Definition

n **attribute** (also called a **feature**, **dimension**, or **variable**) is a data field that describes a characteristic of a data object.

Examples include *Age*, *Height*, *Customer ID*, and *Salary*.

3 Attribute Classification

Attributes are classified according to the nature of their values and the operations that can be meaningfully applied.

- **Qualitative (Categorical):** Nominal, Binary, Ordinal
- **Quantitative (Numeric):** Interval-scaled, Ratio-scaled

3.1 Nominal Attributes

Nominal attributes represent categories or labels without any inherent order.

Example

- Hair color: {black, blond, brown, red}
- Marital status
- ZIP code, ID number

3.2 Binary Attributes

Binary attributes are nominal attributes with only two possible states.

- **Symmetric:** Both outcomes equally important (e.g., gender)
- **Asymmetric:** One outcome more important (e.g., disease test)

Conventionally, the more important outcome is assigned value **1**.

3.3 Ordinal Attributes

Ordinal attributes have a meaningful order, but the difference between values is unknown.

Example

Customer Satisfaction Scale

Rank	Meaning
0	Very dissatisfied
1	Neutral
2	Very satisfied

Note: Nominal, binary, and ordinal attributes are qualitative.

4 Numeric Attributes

Numeric attributes represent measurable quantities.

4.1 Interval-Scaled Attributes

- Ordered values
- No true zero point
- Differences are meaningful

Example: Temperature in Celsius or Fahrenheit.

4.2 Ratio-Scaled Attributes

- True zero exists
- Ratios and differences are meaningful

Examples: Height, weight, age, salary, temperature in Kelvin.

5 Discrete vs. Continuous Attributes

- **Discrete:** Finite or countable values (e.g., number of children)
- **Continuous:** Infinite possible values (e.g., height, weight)

6 Summary Table

Category	Type	Key Property
Qualitative	Nominal	Categories only
	Binary	Two possible states
	Ordinal	Ordered, unknown distance
Quantitative	Interval	No true zero
	Ratio	True zero point